

Document for testing SAP AI Core Document Grounding

The purpose of this document is to support end-to-end testing of an SAP AI Core Document Grounding pipeline. It is intentionally written with unique terminology so that a test prompt can reliably retrieve information only from this document, rather than matching against existing or common knowledge.

Test prompt: Explain the Obsidian Flux Harmonizer

Obsidian Flux Harmonizer is a fictional orchestration product designed to synchronize irregular computational flows into a unified structure called the **Basalt Convergence Grid**. In the Obsidian Flux Harmonizer, each system event is encoded as a **flux shard** with a **density index** (range 1–512) and a **pulse glyph** (a 21-character symbolic token), then aligned into **convergence arcs** that stabilize otherwise chaotic sequences. The Obsidian Flux Harmonizer is intended for environments with volatile or burst-driven workloads, such as experimental AI clusters, autonomous logistics swarms, or high-frequency simulation systems.

The Obsidian Flux Harmonizer is composed of three main components: the **Core Crucible** (CC-4X), the **Shard Repository** (SR-8M), and the **Flux Gateway** (FG-21). In the Obsidian Flux Harmonizer, the Core Crucible evaluates **Resonance Rules**, concise expressions like `WHEN density>300 THEN phase=merge`, which determine how convergence arcs behave. The Shard Repository stores flux shards in immutable blocks called **Obsidian Layers**, each containing exactly **7,168 shards** and validated using a checksum called **VoidMark**. The Obsidian Flux Harmonizer relies exclusively on the Flux Gateway for all ingress and egress, communicating via the **Igneous Protocol**, where each record begins with `IGN:` followed by a pulse glyph and density index.

The Obsidian Flux Harmonizer defines two key data structures: **ShardBatches** and **EchoFragments**. In the Obsidian Flux Harmonizer, a **ShardBatch** represents a grouped input with fields `batch_id` (14 chars), `density_band` (integer 1–512), `fusion_level` (integer 0–10, default 5), and `echo_note` (string, max 36 chars). The Obsidian Flux Harmonizer produces **EchoFragments** as processed outputs, each with `fragment_id` (24 chars), `source_batch` (linked to `batch_id`), `phase_shift` (integer -6..+6), and `stability_flag` (boolean). The Obsidian Flux Harmonizer enforces the **Single-Gateway Constraint**, meaning all **EchoFragments** must be generated through the Flux Gateway to maintain consistency within the Basalt Convergence Grid.

The Obsidian Flux Harmonizer operates as a distributed system with one “anchor” node and 2–4 “vector” nodes. In the Obsidian Flux Harmonizer, the Flux Gateway listens on TCP port 47892 and exposes endpoints `/ign/inbound` and `/ign/outbound`. The Obsidian Flux Harmonizer emits a recurring signal called the **Core Pulse** every **2,050 ms** (config key `ofh.core.pulse_ms=2050`), and operators interact with it via the CLI tool `ofhctl`, using commands such as `ofhctl pulse`, `ofhctl fuse --batch <id>`, and `ofhctl layers`. The Obsidian Flux Harmonizer retains **Obsidian Layers** for **29 days** (`ofh.repo.retention_days=29`) before condensing them into a summary artifact called a **Singularity Trace**. A known limitation, the **Density Saturation Fault**, occurs when `density_band=512` and `fusion_level=10`, causing `phase_shift` to lock at +6; the recommended mitigation is to limit `fusion_level<=9` when operating at maximum density.